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The Editor Journal of Flood Risk Management

Dear Editor,

I am pleased to submit the manuscript “When do flood susceptibility models beat the flood record? Evidence from radar-observed floods in Bangladesh” for consideration as an Original Paper in the Journal of Flood Risk Management.

Every flood manager in a radar-mapped region already holds a free flood map, the record of where floods have been, yet published flood susceptibility maps are almost never tested against it. Weather and river forecasts must beat persistence before their skill is believed, while susceptibility studies in Bangladesh report areas under the ROC curve (AUC) of 0.87 to 0.98, usually from random splits of spatially clustered points that test a model beside its own training data. The manuscript sets the model against the record on later floods, using flood extents mapped from Sentinel-1 radar and scoring every model on 10 km blocks held out from training, and asks when the record is enough and when a model adds to it.

The main findings are these.

- Predicting each flood year from the years before it, in the three study regions and four validation regions that share no asset with them, gives 15 tests. The record’s lead over a gradient-boosted terrain model fell as the share of a flood reaching ground no earlier flood had reached rose (Spearman $\rho = -0.775$, $p < 0.001$), whatever the length of the record, and the two were level at about 33 % of flooded assets on such new ground (95 % interval 18–41 %, or 3–43 % when whole regions are resampled). The relation held with each region left out in turn and within every region with more than one test, so the answer depends on the flood regime rather than resting on one region or one year.
- In Rangpur and Rajshahi, where few of the 2024 floods’ assets stood on new ground, the record ranked the flooding at 0.896 against 0.801 for the model and kept its lead under every check the manuscript applies, including masks re-mapped from a single satellite and wider validation blocks. On the cyclone-affected coast, where most flooding reached new ground, the model led. Given the record as a feature, the model beat both of its parts in 5 of 8 tests, all riverine.
- A global hydrodynamic hazard map (JRC, LISFLOOD-FP), a fair rival only for river flooding, was no stronger than the model, trailing the record significantly in 5 of 15 tests and leading it in 4, all on floods that reached much new ground. The comparison thus bears on hazard maps generally, as it has long done in earthquake forecasting, where maps of past seismicity remain the reference that forecasts must beat.
- Training on radar-observed extents rather than terrain-threshold labels raised the AUC by 0.080 to 0.101 on held-out areas, and by 0.024 to 0.071 against the 2024 floods, which neither label source had seen.

I believe the paper suits the Journal because its conclusions concern how flood risk is managed as much as how it is modelled. It recommends that agencies publish the radar flood record as a planning baseline wherever one exists, that the choice between that record and a model follow how much of past floods fell on new ground, and that maps used to site shelters, embankments or schools be judged on held-out areas and later floods. It also engages the Journal’s own work on flood hazard mapping in Bangladesh, radar flood mapping, infrastructure exposure and vulnerability.

The manuscript states its limits plainly. The radar flood maps agree only moderately with an independent MODIS flood map (critical success index 0.29 and 0.34), so every score measures agreement with flooding as radar sees it, 10 of the 15 tests cannot separate the record from the model, and the crossover near a third of flooding on new ground is estimated from seven regions of one country. Every number in the text is generated directly from the pipeline outputs, the code is openly available at https://github.com/rakibhhridoy/rimes-mapathon, and the input data and outputs are archived at https://doi.org/10.5281/zenodo.23573843. A Supporting Information file gives the pipeline settings, the robustness checks, the graph network and tabular model comparison, and the region-level analysis of the tests.

The main text runs to just under 6,000 words, excluding references, figures and tables, with 5 figures and 7 tables.

The manuscript is original, has not been published, and is not under consideration elsewhere. The open-source software used here is described in a separate manuscript under review at SoftwareX (SOFTX-D-26-01375), which reports the software and its validation design and applies it across the whole country; this manuscript reports the scientific findings for the three study regions. The software paper repeats a few of the validation figures given here, as examples of what the software produces, but does not analyse them. I am its sole author and approve its submission. I declare an affiliation with Fermium Systems, which hosts the public dashboard described in the paper. The software is released under the MIT licence and the data under open licences, and no result depends on a proprietary component. The work received no specific funding.

Suggested reviewers:

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Thank you for considering the manuscript.

Yours sincerely,

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